Assignment 8 — Basic SQL Queries

Project: College Management System | Database: college_management_system | SQL dialect: MySQL 8.0

Based on the seven-table college design supplied for this project. Example records and expected outputs are illustrative, not screenshots from MySQL Workbench.

Queries, explanations and expected outputs

The following results are computed from the provided fictional sample data; actual MySQL Workbench screenshots must be captured after execution.

1. WHERE: students in Computer Applications

```
SELECT Student_ID, Name, Admission_Year FROM Student WHERE Department_ID=1 ORDER BY
Student_ID;
```

Expected output from the included demonstration dataset (3 rows):

Student_ID	Name	Admission_Year
101	Aarav Sharma	2025
102	Diya Nair	2025
105	Rohan Das	2024

2. ORDER BY: courses by credits

```
SELECT Course_Name, Credits FROM Course ORDER BY Credits DESC, Course_Name;
```

Expected output from the included demonstration dataset (5 rows):

Course_Name	Credits
Database Management Systems	4
Probability and Statistics	4
Python Programming	4
Accounting Principles	3
Computer Networks	3

3. GROUP BY and COUNT: students per department

```
SELECT d.Department_Name, COUNT(s.Student_ID) AS Students FROM Department d LEFT JOIN
Student s ON s.Department_ID=d.Department_ID GROUP BY d.Department_ID,d.Department_Name
ORDER BY d.Department_ID;
```

Expected output from the included demonstration dataset (3 rows):

Department_Name	Students
Computer Applications	3
Commerce	2
Mathematics	1

4. HAVING: courses with 2 or more enrollments

```
SELECT c.Course_Name, COUNT(*) AS Enrollments FROM Enrollment e JOIN Course c ON
c.Course_ID=e.Course_ID GROUP BY c.Course_ID,c.Course_Name HAVING COUNT(*)>=2 ORDER BY
Enrollments DESC,c.Course_Name;
```

Expected output from the included demonstration dataset (3 rows):

Course_Name	Enrollments
Database Management Systems	3
Accounting Principles	2
Python Programming	2

5. SUM: credits registered by student

```
SELECT s.Name, SUM(c.Credits) AS Total_Credits FROM Enrollment e JOIN Student s ON
s.Student_ID=e.Student_ID JOIN Course c ON c.Course_ID=e.Course_ID GROUP BY s.Student_ID,
s.Name ORDER BY s.Student_ID;
```

Expected output from the included demonstration dataset (6 rows):

Name	Total_Credits
Aarav Sharma	8
Diya Nair	7
Kabir Singh	3
Ananya Iyer	4
Rohan Das	8
Sara Khan	3

6. AVG MIN MAX: examination marks

```
SELECT ROUND(AVG(Marks),2) AS Average_Marks, MIN(Marks) AS Minimum_Marks, MAX(Marks) AS
Maximum_Marks FROM Result;
```

Expected output from the included demonstration dataset (1 rows):

Average_Marks	Minimum_Marks	Maximum_Marks
81.89	69	95

7. Manipulate data safely: read-only projection

```
SELECT Name, Email FROM Student WHERE Admission_Year=2025 ORDER BY Name;
```

Expected output from the included demonstration dataset (4 rows):

Name	Email
Aarav Sharma	aarav@college.edu
Ananya Iyer	ananya@college.edu
Diya Nair	diya@college.edu
Sara Khan	sara@college.edu